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Plasma cell-free DNA methylation-based prognosis in metastatic castrate-resistant prostate cancer



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Molecular prognostication in metastatic castration-resistant prostate cancer (mCRPC) remains a challenge due to the lack of validated biomarkers. This study developed a plasma cell-free DNA (cfDNA) methylation-based prognostic model in mCRPC. Targeted cfDNA methylation sequencing in 96 prostate cancer patients across different states of cancer progression revealed 77 methylation haplotype blocks (MHBs) differentially methylated from organ-confined prostate cancer to mCRPC states. Among these 77 MHBs, the top 20 MHBs were associated with mCRPC overall survival and most MHB methylation levels positively correlated with predicted circulating tumor DNA (ctDNA) fraction. By integrating the MHB-based risk score with currently available prognostic clinical variables and ctDNA fraction, a prognostic nomogram was developed which showed high predictive performance for mCRPC survival (AUC = 0.99 for 6 months, AUC = 0.84 for 1 year, and AUC = 0.82 for 2 years). These findings demonstrate potential of cfDNA methylation as a molecular biology-driven biomarker for mCRPC prognosis.

Prostate cancer is the second most common cancer in men globally and a leading cause of cancer-related deaths, with over 396,000 deaths annually worldwide¹. Despite significant advancements in therapeutic strategies, survival for metastatic castration-resistant prostate cancer (mCRPC) remains poor, with a median survival time of 30 to 34 months², underscoring the need for improved prognostication methods. Since 2015, the management of metastatic hormone-sensitive prostate cancer (mHSPC) has rapidly evolved, with androgen deprivation therapy (ADT)-based combinations emerging as the new standard of care³. Several of these novel treatments target the androgen receptor-testosterone pathway, utilizing agents such as Androgen Receptor Pathway Inhibitors that were initially developed for mCRPC. Despite the rapid adoption of these intensified mHSPC drug combinations, progression to mCRPC is inevitable.

Currently, prognostication in mCRPC relies primarily on clinical variables, either with circulating tumor cell number (>5 per 7.5 milliliters blood)⁴ or with non-specific blood-based proteins such as alkaline phosphatase (ALP), lactate dehydrogenase (LDH), albumin, and hemoglobin

levels^{5–7}. Clinical factors such as the Eastern Cooperative Oncology Group (ECOG) performance status, opioid use for pain management, and presence of visceral metastases are also included to derive prognostic risk groups in mCRPC². While these markers offer the advantages of being validated in prospective randomized clinical trials and are easily measured in blood samples or clinical history, they are not mCRPC tumor biology-specific. Consequently, it is challenging to leverage prognostic risk groups developed from these markers to develop targeted therapeutic interventions in mCRPC patients with poor prognostic risk.

To address these limitations, efforts to integrate blood molecular profiling with the non-specific blood-based biomarkers into prognostication have emerged in the last decade. Plasma cell-free DNA (cfDNA) alterations, circulating tumor DNA (ctDNA), and specific genomic aberrations such as *TP53* mutations, *AR* amplifications, and *RBI* loss have shown promise as prognostic classifiers in metastatic prostate cancer^{8–13}. While many of these genomic alterations are truncal and present in the primary tumor, their detection in ctDNA can be limited by low tumor DNA

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Table 1 | Clinical characteristics of patients in different states of prostate cancer progression

	Localized PC	mHSPC	mCRPC
Total Patients	19	28	49
Age at sample collection (years)	68 (57–85)	73.5 (58–85)	70 (53–86)
PSA at sample collection (ng/mL)	2.9 (0–38.1)	9.15 (0.01–1401.8)	7.5 (0.1–1863.6)
ALP at sample collection (U/L)	69 (44–492)	77 (47–987)	88 (41–639)
LDH at sample collection (U/L)	198 (150–276)	202 (131–442)	217 (122–811)
Hemoglobin at sample collection	14.6 (7.8–18)	15.05 (12.4–18.1)	13.1 (8.6–16)
Albumin at time of sample collection	4.1 (3.4–4.8)	4.05 (3.6–4.6)	4 (3.5–4.7)
Metastatic Volume at mHSPC			
Low	0	13	12
High	0	2	11
Unknown	19	13	26
Gleason Score at ID			
≤7	9	11	18
≥8	10	16	31
Clinical T Staging at ID/Resection			
T1c	4	2	2
T2	4	4	2
T2a	1	1	4
T2b	2	1	2
T2c	0	3	6
T3	1	1	2
T3a	3	2	3
T3b	4	4	3
T4	0	3	3
TX	0	7	19
T-unknown	0	0	3
Median mHSPC follow-up time from sample collection to last follow-up (months)	NA	42.25 (13.34–46.92)	NA
Median time from first mHSPC treatment to failure (months)	NA	10.53 (3.2–42.23)	13.07 (3.1–82.03)
mHSPC Patients progressed on ADT by date of analysis	NA	7	NA
Median mCRPC follow-up time from sample collection to last follow-up (months)	NA	NA	27.11 (1.94–46.92)
Median follow-up time from date of mCRPC to last follow-up (months)	NA	NA	50.2 (11.3–199.37)
Patients deceased by date of analysis	1	5	34

PC prostate cancer, mHSPC metastatic hormone sensitive prostate cancer, mCRPC metastatic castration resistant prostate cancer, PSA prostate specific antigen, ALP alkaline phosphatase, LDH lactate dehydrogenase, ID initial diagnosis.

^aData are median (range).

shedding in localized prostate cancer^{14,15}. Even within the mCRPC state, clonal evolution and tumor heterogeneity can make genomic profiling challenging¹⁶. In contrast, DNA methylation has emerged as a promising biomarker due to its enhanced sensitivity and stability^{17,18}. It is well known that DNA methylation changes arise early in tumorigenesis and span broad genomics regions that can be detected even at low tumor fractions¹⁹. Additionally, DNA methylation patterns are tissue-specific which provide insights into tissue of origin and improve cancer discrimination²⁰.

In this study, we examined plasma cfDNA methylation profiles in 96 prostate cancer patients from the localized, organ-confined states to the metastatic states of prostate cancer progression. We employed methylation haplotype blocks (MHBs) to identify fragment level co-methylation signatures specific to the state of prostate cancer progression and then developed epigenetic prognostic classifiers for mCRPC lethality by integrating clinical and epigenetic markers. Our goal was to establish a clinical-molecular tool specific to mCRPC-biology that enhances prognostication and potentially guides personalized targeted treatment strategies based on molecular pathways associated with the aggressive mCRPC state.

Results

Clinical characteristics of prostate cancer patients

We enrolled 96 prostate cancer patients after obtaining written informed consent with adequate longitudinal follow up for mCRPC patients. Details of the real-world database are provided under Supplementary Methods. Briefly, the patient cohort consisted of individuals with localized prostate cancer ($n = 19$), mHSPC ($n = 28$), and mCRPC ($n = 49$). Clinical variables were collected from Electronic Medical Records (EMRs) and cohort demographics are summarized in Table 1. All 28 mHSPC patients provided blood samples prior to any androgen deprivation treatment. The mean yield of cfDNA was 18.96 nanograms of cfDNA per milliliter of plasma (ranging 2–612 ng/mL) (Supplementary Data 1).

Mapping characteristics of targeted methylation sequencing

We performed Enzymatic Methyl-Seq (EM-Seq) to identify 5-methylcytosine (5-mC) at CpG sites in 437 target regions selected from peer-reviewed studies, TCGA data, and prostate cancer related oncogene and tumor-suppressor gene regions (Supplementary Data 2). Across patient

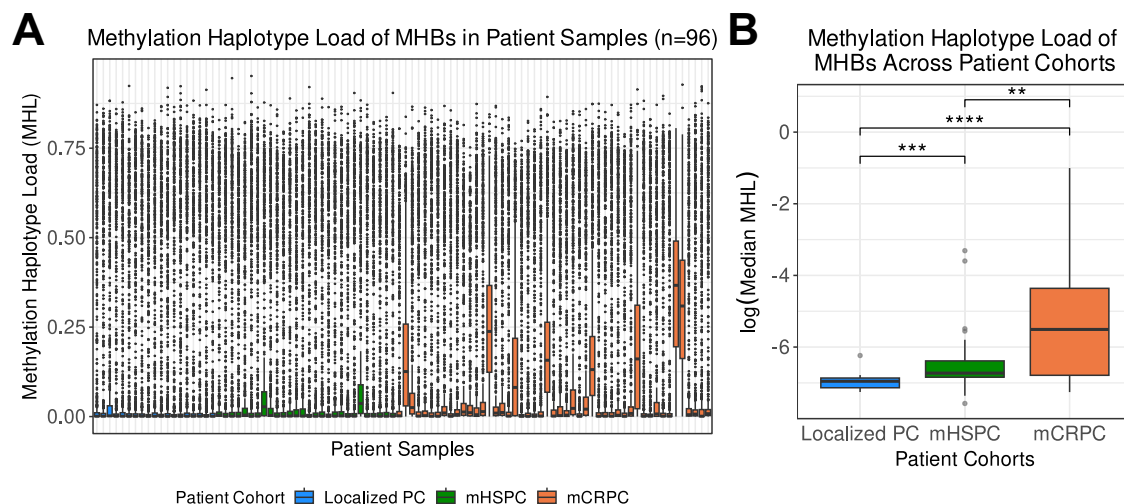


Fig. 1 | Methylation haplotype load increases with the advancement of prostate cancer. **A** Box plot depicting the MHL for each MHB ($n = 1194$) across the patients representing localized prostate cancer (blue, $n = 19$), mHSPC (green, $n = 28$), and mCRPC (orange, $n = 49$). **B** Box plot showing the distribution of median MHL across patient cohorts. Statistical significance between cohorts assessed with

Wilcoxon rank-sum test (** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.001$). MHL methylation haplotype load, MHB methylation haplotype block, PC prostate cancer, mHSPC metastatic hormone sensitive prostate cancer, mCRPC metastatic castration resistant prostate cancer.

samples, we achieved a mean mapping efficiency of 77.4% (ranging 48.4–80.3%). Among mappable reads, 60.5% (range: 41–68%) were aligned to the target regions, and 1.9% (range: 1.1–2.7%) were duplicate alignments (Supplementary Fig. 1). Among the unique alignments in the target regions, we found that 99.3% had coverage greater than 10x and overall mean coverage was 96x (Supplementary Fig. 2).

Distribution of methylation haplotype blocks

Following alignment and methylation calling with Bismark²¹, we utilized mHapSuite²² to conduct an exploration of MHBs, which we defined as genomic regions containing at least three bases and a linkage disequilibrium value (r^2) greater than 0.3, across patients representing different states of prostate cancer. This analysis identified a total of 3525 MHBs in the 437 target regions across all patient samples. After we excluded MHBs with less than 3 CpGs and median read counts below 50, 1530 MHBs remained for further analysis. Among these, 130 MHBs were located within positive control regions, while 16 MHBs were within negative control regions (Supplementary Fig. 3). The median methylation was 84.6% among the positive control regions and 0.3% among the negative control regions (Supplementary Fig. 4).

Among the 437 target regions, 366 were involved in genes implicated in cancers, of which 263 were linked to prostate cancer based on prior peer-reviewed studies and known oncogenes and tumor suppressor genes (Supplementary Data 2). These 366 cancer-associated regions were represented by 1194 MHBs (Supplementary Fig. 3). The mean length of these MHBs was 127 base pairs (bp) (ranging from 5 to 2100 bp) and the mean number of CpGs per MHB was 11 CpGs (ranging from 3 to 137 CpGs) (Supplementary Fig. 5). Since MHBs are defined based on linkage disequilibrium and span contiguous genomic regions, they can overlap multiple functional elements and be categorized into more than one CpG or genic feature. The 1194 MHBs were primarily located in CpG islands (76.3% of MHBs) and CpG shores (30% of MHBs). Furthermore, these regions largely consisted of promoters (47.3% of MHBs), introns (68.6% of MHBs), exons (50.3% of MHBs), and genic regions 1 to 5 Kb from transcription start sites (TSS) (53.2% of MHBs) (Supplementary Fig. 6).

Differential methylation haplotype analysis reveals progressive methylation changes throughout progression of prostate cancer

To investigate methylation changes across progressive prostate cancer states, we calculated successive CpG methylation at varying fragment

lengths by determining the methylation haplotype load (MHL) across the 1194 MHBs (Supplementary Data 3)^{22,23}. This analysis demonstrated increasing MHL values as the disease progressed from localized organ-confined prostate cancer to the metastatic states (Fig. 1A, B). To identify differentially methylated regions (DMRs) among localized prostate cancer, mHSPC, and mCRPC states, we excluded any MHBs with missing values ($n = 71$ MHBs). From the remaining 1123 MHBs, we partitioned 203 MHBs with $MHL > 0.05$ in localized prostate cancer for separate analysis. The rationale behind the filtering criteria is the low ctDNA fraction found in localized prostate cancer^{14,15} which could lead to low cancer-specific methylation detection in the localized state. High methylation level found at the localized state may represent contributions from non-tumor cells such as lymphocytes. Therefore, the remaining 920 MHBs with low successive methylation load ($MHL \leq 0.05$) were used in subsequent analyses (Fig. 2A).

From the 920 MHBs, we identified 81 DMRs across localized prostate cancer and mHSPC (two-tailed Welch's unpaired t-test, $p < 0.05$) (Fig. 2B, Supplementary Data 4). When comparing mHSPC with mCRPC, we identified 699 DMRs (two-tailed Welch's unpaired t-test, $p < 0.05$), of which 77 DMRs were found in common with the analysis comparing localized prostate cancer with mHSPC (two-tailed Welch's unpaired t-test, $p < 0.05$, false discovery rate (FDR) < 0.1) (Fig. 2B, C, Supplementary Data 5). These DMRs all demonstrated a consistent positive shift in MHL values as the disease progressed, suggesting progressive hypermethylation within target regions.

Target region methylation and clinical biomarkers are associated with overall survival

To investigate the prognostic potential of the identified DMRs with overall survival in mCRPC, we selected the top 20 of the 77 common DMRs identified in the comparative analysis between localized, mHSPC, and mCRPC states for further evaluation. Among the 49 mCRPC patients, 34 died during follow-up and had a median survival time of 17.5 months (Supplementary Data 1). Univariate Cox proportional hazards survival analyses showed that all selected 20 DMRs were significantly associated with overall survival ($p < 0.05$) (Table 2). To reduce redundancy and avoid overrepresentation of signals from a single gene, we excluded DMRs associated with the same gene, as these regions were highly correlated (Supplementary Fig. 7). The resulting 15 DMRs were used to generate an MHB-based composite prognostic risk score for mCRPC survival analysis. A univariate Cox analysis of the MHB composite risk score showed significant

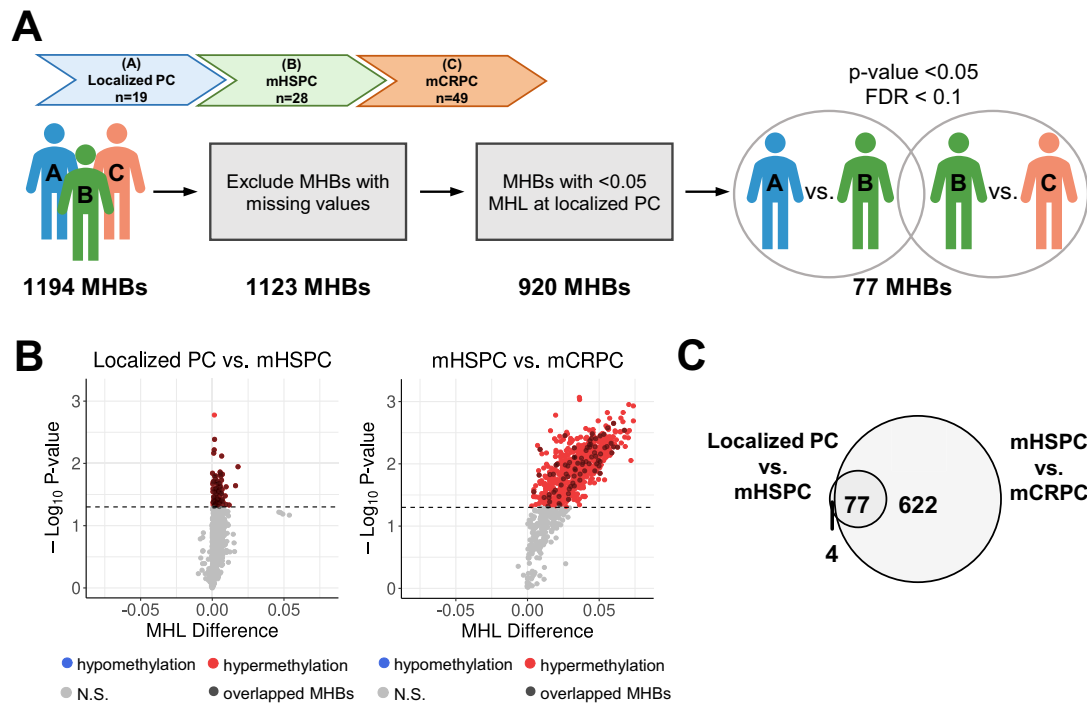


Fig. 2 | MHBs are differentially methylated across the states of prostate cancer.

A Overview of the pipeline to identify MHBs and then perform comparative analyses between the progressive states of prostate cancer. **B** Volcano plots demonstrating the differentially methylated MHBs found when comparing localized PC to mHSPC and mHSPC to mCRPC. Hypomethylated MHBs (MHL difference < 0) are in blue, hypermethylated MHBs (MHL difference > 0) are in red, and MHBs shared across all states are found in black. The reference group for hypo- or hyper-methylated

MHBs is localized PC in the localized PC vs. mHSPC comparison, and mHSPC in the mHSPC vs. mCRPC comparison. **C** Venn diagram depicting the significant MHBs ($p < 0.05$) which are overlapping between the comparative analyses. *MHL* methylation haplotype load, *MHBs* methylation haplotype blocks, *PC* prostate cancer, *mHSPC* metastatic hormone sensitive prostate cancer, *mCRPC* metastatic castration resistant prostate cancer.

Table 2 | Significant association of top 20 MHBs with mCRPC survival

Gene	MHB Region	p value	HR (95% CI)
ALOX5	chr10:45418808–45420022	2.34E–07	2.27 (1.66–3.09)
HIC1	chr17:2056975–2057271	1.96E–07	2.45 (1.75–3.43)
HIC1	chr17:2058035–2058073	1.10E–07	2.66 (1.85–3.82)
TMEM106A	chr17:43211484–43212450	2.52E–08	2.82 (1.96–4.06)
HOXB5	chr17:48596419–48596858	1.74E–07	2.48 (1.76–3.49)
cg15222899	chr18:57427748–57428143	1.51E–07	2.32 (1.69–3.18)
WFDC2	chr20:45469884–45470211	1.43E–07	2.47 (1.76–3.45)
EFEMP1	chr2:55923171–55923364	1.91E–06	2.27 (1.62–3.18)
SOX11	chr2:5691072–5691083	1.23E–07	3.18 (2.07–4.89)
cg09064304	chr4:140498280–140498336	2.07E–08	3.29 (2.17–4.98)
cg09064304	chr4:140498337–140498442	1.40E–08	2.97 (2.04–4.33)
SCGB3A1	chr5:180591347–180591432	7.47E–08	2.63 (1.85–3.75)
SCGB3A1	chr5:180591433–180591585	5.90E–08	2.57 (1.83–3.62)
SCGB3A1	chr5:180591586–180591782	1.12E–07	2.58 (1.82–3.66)
FAM115A	chr7:143885367–143885557	2.80E–07	2.63 (1.82–3.8)
KCNH2	chr7:150958529–150958549	8.07E–08	3 (2.01–4.49)
PTPRN2	chr7:157686104–157686172	7.04E–08	2.44 (1.76–3.38)
PTPRN2	chr7:157691808–157692130	6.90E–08	2.45 (1.77–3.39)
PRDM14	chr8:70070419–70070468	6.16E–08	2.46 (1.78–3.41)
CDKN2A	chr9:21989731–21989796	4.43E–05	1.87 (1.39–2.53)

MHBs methylation haplotype blocks, HR Hazard Ratio, CI Confidence Interval, mCRPC metastatic castration resistant prostate cancer.

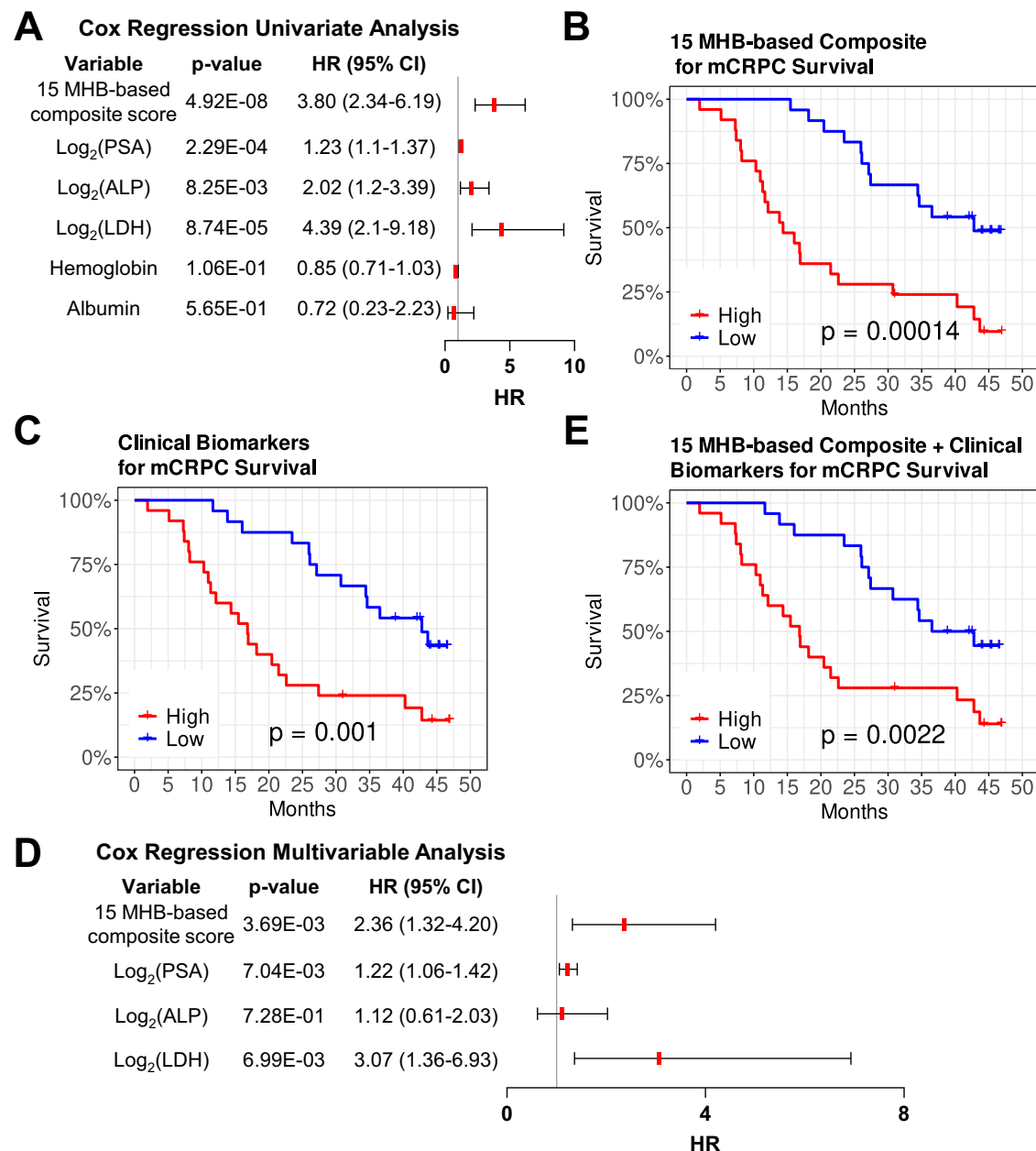


Fig. 3 | Top MHBs and clinical biomarkers are associated with mCRPC survival. **A** Forest plot of HRs and 95% CI from univariate Cox analyses. **B** Kaplan–Meier plot of survival in mCRPC patients stratified into high-risk and low-risk groups based on a 15 MHB composite score (log-rank test, $p = 0.00014$). **C** Kaplan–Meier plot of high- and low-risk mCRPC survival groups based on a multivariable composite score of PSA, ALP, and LDH (log-rank test, $p = 0.001$). **D** Forest plot of HRs and 95%

CI derived from multivariable Cox analysis. **E** Kaplan–Meier plot of high- and low-risk mCRPC survival groups based on the multivariable composite risk score combining the 15 MHB composite score and clinical biomarkers (log-rank test, $p = 0.0022$). *MHB* methylation haplotype block, *mCRPC* metastatic castration resistant prostate cancer, *PSA* prostate specific antigen, *ALP* alkaline phosphatase, *LDH* lactate dehydrogenase, *HR* hazard ratio, *CI* confidence interval.

association with mCRPC survival ($p < 0.05$, hazard ratio (HR) = 3.8) (Fig. 3A). Patients with risk scores above the median risk score were classified as high-risk, while those with risk scores below the median were classified as low-risk. We found that mCRPC patients with high-risk scores had significantly worse overall survival. Specifically, the median survival time of high-risk mCRPC patients was 14.4 months while the median survival time of low-risk patients was 42.8 months during the 47-month follow-up period (log-rank test, $p = 0.00014$) (Fig. 3B).

When examining the prognostic potential of the known clinical biomarkers (PSA, LDH, ALP, albumin, and hemoglobin), univariate Cox analyses showed that increased PSA, LDH, and ALP levels were associated with worse overall survival in mCRPC patients ($p < 0.05$) (Fig. 3A). In a

multivariable Cox model combining PSA, LDH, and ALP, the median survival times were 16.8 months for high-risk patients and 42.8 months for low-risk patients (log-rank test, $p = 0.001$) (Fig. 3C).

To assess the potential to enhance mCRPC prognostication, the MHB-based composite score was integrated with prognostic clinical biomarkers. Upon multivariable analysis, integrating the MHB-based composite score with significant clinical prognostic factors from univariate Cox analyses (PSA, LDH, ALP), the MHB-based score remained significantly associated with mCRPC survival ($p = 0.00369$) (Fig. 3D). This multivariable model demonstrated that high-risk mCRPC patients had significantly worse overall survival than low-risk patients, with median survival times of 16.8 vs. 39.7 months respectively (log-rank test, $p = 0.0022$). (Fig. 3E).

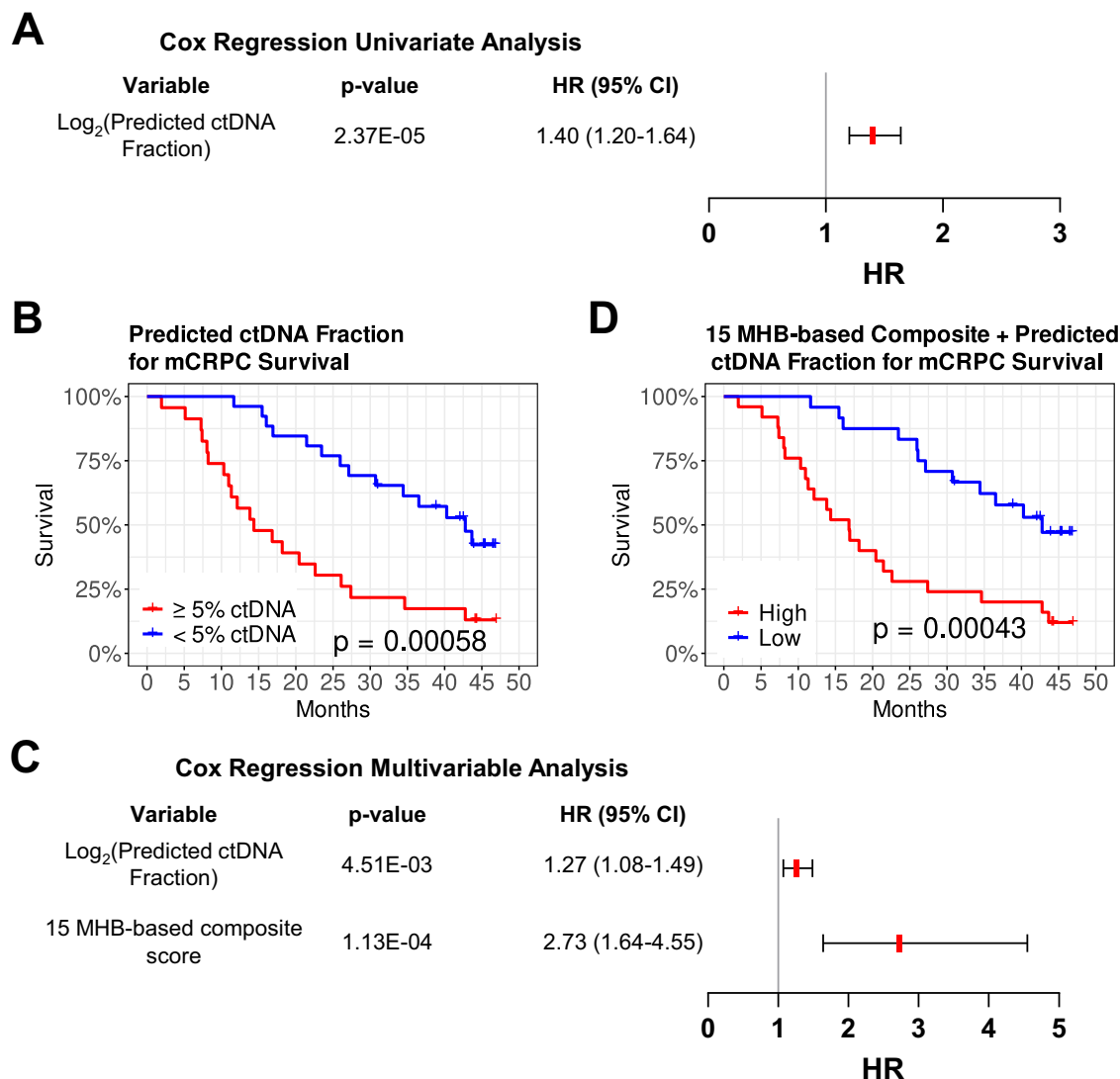


Fig. 4 | ctDNA fraction and top MHBs are associated with mCRPC survival. **A** Forest plot of HR and 95% CI from univariate Cox analysis of predicted ctDNA fraction. **B** Kaplan–Meier showing mCRPC survival probability based on high ($\geq 5\%$) and low ($\leq 5\%$) ctDNA fraction (log-rank test, $p = 0.00058$). **C** Forest plot of HRs and 95% CI derived from multivariable Cox analysis. **D** Kaplan–Meier plot

showing mCRPC survival in high-risk and low-risk patient groups based on a multivariable composite risk score combining predicted ctDNA fraction and the 15 MHB composite score (log-rank test, $p = 0.00043$). MHB methylation haplotype block, mCRPC metastatic castration resistant prostate cancer, HR hazard ratio, CI confidence interval.

Nontumor-derived methylation regions also show prognostic potential

In addition to MHBs with low methylation, we also examined the 203 MHBs with high successive methylation at localized prostate cancers ($\text{MHL} > 0.05$). Although these methylation signals were unlikely derived from tumor cells, the methylation changes in cfDNA could reflect tumor microenvironment and immune responses. Comparative analysis identified 37 DMRs between localized prostate cancer and mHSPC, of which 5 remained significant when comparing mHSPC with mCRPC (two-tailed Welch's unpaired t-test, $p < 0.05$) (Supplementary Fig. 8). We further examined the prognostic potential of these MHBs and observed that 4 of the 5 MHBs were significantly associated with mCRPC survival in univariate Cox analysis ($p < 0.05$) (Supplementary Data 6). The MHB-based composite risk score based on the top 3 DMRs associated with unique gene regions also significantly distinguished mCRPC patients with different survival risks (Supplementary Fig. 9A). Combining the 3-MHB composite score with significant clinical prognostic factors also demonstrated the MHB-based score remained significantly

associated with mCRPC survival and the multivariable model stratified high-risk and low-risk mCRPC patients (Supplementary Fig. 9B, C).

Methylation markers correlate with ctDNA fraction

To explore whether ctDNA fraction might influence survival predictions in this patient cohort, we utilized the online ctDNA fraction prediction tool in mCRPC state (<https://www.ctdna.org>)²⁴. Predicted ctDNA fraction of patient samples was performed using the concentrations of plasma cfDNA, PSA, LDH, and ALP (Supplementary Data 1). Considering that ctDNA fraction can reflect tumor burden and disease progression^{25–27}, we assessed whether its variation among patients would be reflected in survival predictions. In line with previous studies in mCRPC^{12,28,29} demonstrating that high ctDNA fraction was observed to be associated with poor prognosis, we observed that the estimated ctDNA fraction was highly associated with mCRPC survival ($p < 0.0001$) (Fig. 4A). Patients with low ctDNA fraction ($< 5\%$) had a median survival time of 42.8 months and patients with high ctDNA fraction ($\geq 5\%$) had a median survival time of 14.4 months (log-rank test, $p = 0.00058$) (Fig. 4B). We further evaluated whether MHL values in the

MHBs correlated with the predicted ctDNA fraction. Among the MHBs which were significantly associated with mCRPC survival (Cox $p < 0.05$) and patients with ctDNA fraction at or above 1% ($n = 35$), we found that the MHL values in 780 MHBs positively correlate with ctDNA fraction (Pearson $r > 0.5$). (Supplementary Data 7, Supplementary Fig. 10A). Interestingly beyond these MHBs, 12 MHBs were not correlated (Pearson $-0.2 < r < 0.2$) and 43 MHBs were inversely correlated (Pearson $r < 0$) with ctDNA fraction (Supplementary Data 7, Supplementary Fig. 10B, C). While the positively correlated and non-correlated MHBs have low MHL (< 0.05) at localized PC, the inversely correlated MHBs have high MHL (> 0.05) at localized PC (Supplementary Data 7).

cfDNA methylation is an independent predictor for survival in a multi-modal risk model

Clinical variables, ctDNA fraction, and cfDNA methylation observed to be associated with survival were integrated into a single multi-modal model for prognostic performance. We first combined the ctDNA fraction and the 15 MHB-based composite score into a multivariable composite risk score. Multivariable analysis demonstrated that both factors were independently associated with overall survival (Fig. 4C). A nested model comparison using ANOVA showed that adding the 15 MHB-based composite score significantly improved fit over ctDNA alone (Likelihood ratio test = 15.451, p -value = 0.00085). Of note, in a correlation analysis between the 15 MHB-based composite score and ctDNA fraction there were 8 outlier cases which exhibited poor correlation (residual $> \pm 1$) suggesting that the MHB-based score may capture prognostic information not reflected by ctDNA alone (Supplementary Fig. 11). Together, the combined model efficiently stratified low-risk and high-risk patients, with a median survival time of 42.8 months in low-risk patients and 16.8 months in high-risk patients (log-rank test, $p = 0.00043$) (Fig. 4D). We then incorporated the clinical prognostic biomarkers of PSA, LDH, and ALP into the multivariable composite score to establish a multi-modal risk score. Among all survival-associated variables, the 15 MHB-based composite score remained significantly predictive of mCRPC survival ($p = 0.00318$) (Supplementary Fig. 12A). By integrating ctDNA fraction, methylation, and clinical biomarkers, the multi-modal risk score classified survival with the median survival time for the low-risk and the high-risk groups observed to be 42.8 months and 15.5 months, respectively (log-rank test, $p = 0.00025$) (Supplementary Fig. 12B). These findings suggest that methylation markers correlated with ctDNA fraction hold potential as complementary biomarkers for survival prediction in mCRPC.

Prognostic multi-modal nomogram for mCRPC survival

Going beyond risk-group categorizations, we generated nomograms from multi-modal models to develop individualized patient-centric estimations of mCRPC survival. Clinical biomarkers (PSA, LDH, and ALP) associated with mCRPC survival at the univariate level ($p < 0.05$) (Fig. 3B) were included in a multivariable analysis. Only PSA and LDH (at the time of developing castrate resistance) were observed to remain significantly associated with survival ($p < 0.001$) (Supplementary Fig. 12C) and were included in a prognostic nomogram. The performance of the clinical biomarker-based nomogram model was assessed with time-dependent receiver operating characteristic (ROC) analysis with area under the curve (AUC) values of 0.985 ± 0.038 for 0.5-year survival, 0.855 ± 0.172 for 1-year survival, and 0.848 ± 0.134 for 2-year survival (Fig. 5A). The corresponding nomogram utilizing clinical biomarkers alone to predict survival at these time points is presented in Fig. 5B. Meanwhile the performance of a nomogram model utilizing only the 15 MHB-based composite score demonstrated improved range of AUC values at 1-year and 2-year survival with AUC values of 0.971 ± 0.058 for 0.5-year survival, 0.905 ± 0.114 for 1-year survival, and 0.856 ± 0.121 for 2-year survival (Fig. 5C, D).

Integrating the 15MHB-based composite score with clinical biomarkers into a multivariable model (Fig. 3D) showed that PSA, LDH, and the MHB score were associated with mCRPC survival and the model had AUC values of 0.991 ± 0.031 for 0.5-year survival, 0.871 ± 0.165 for 1-year

survival, and 0.848 ± 0.133 for 2-year survival (Supplementary Fig. 12D). Furthermore, combining the 15 MHB-based composite score and clinical biomarkers with the predicted ctDNA fraction into a multivariable model, the 15 MHB-based composite score ($p = 0.00318$, HR = 2.38 (1.34–4.22)) was the only variable significantly associated with survival (Supplementary Fig. 12A). The integrated clinical biomarker, 15 MHB-based composite score and ctDNA fraction nomogram for predicting 0.5-year, 1-year, and 2-year survival showed a higher predictive performance for 0.5-year survival as compared to the nomograms utilizing clinical biomarkers or the 15MHB-based score alone with AUC values of 0.989 ± 0.034 for 0.5-year survival, 0.844 ± 0.178 for 1-year survival, and 0.823 ± 0.144 for 2-year survival (Fig. 5E, F).

Functional enrichment analysis reveals pathways associated with prostate cancer progression

To investigate the potential biological significance of the 77 DMRs identified across the stages of prostate cancer, we performed functional enrichment using GREAT (great.stanford.edu)³⁰. This analysis revealed a significant enrichment for the Gene Ontology (GO) biological process regulation of transmembrane receptor protein serine/threonine kinase signaling pathway (GO:0090092; $p = 4.27\text{E-}7$, FDR = 5.62E-3).

To further dissect the epigenetic landscape associated with late-stage metastatic prostate cancer, we focused on the 272 most significant DMRs (two-tailed Welch's unpaired t-test, $p < 0.01$) from the 622 DMRs identified (Fig. 2C) as uniquely associated with late-stage disease. GREAT analysis of these top mCRPC-associated DMRs revealed enrichment for the GO biological process cell fate commitment (GO:0045165; $p = 1.19\text{E-}0$, FDR = 1.56E-5). Additionally, we observed significant enrichment for the GO molecular functions transcriptional repressor activity, RNA polymerase II core promoter proximal region sequence-specific binding (GO:0001078; $p = 1.07\text{E-}9$, FDR = 4.51E-6) and steroid hormone receptor activity (GO:0003707; $p = 7.02\text{E-}8$, FDR = 1.48E-3).

Discussion

In this study, we profiled tumor-specific cfDNA methylation patterns to identify the DMRs that provide insights into prostate cancer progression, survival prognosis, and risk stratification. The findings reveal that a set of MHBs undergo epigenetic changes across the prostate cancer states and in advanced prostate cancer states. The enrichment of the serine/threonine kinase signaling pathway from the DMRs identified across prostate cancer states suggests that the DMRs may contribute to key mechanisms underlying disease progression as this is a pathway implicated in prostate cancer proliferation, metastasis, and treatment resistance outcomes^{31–34}. The enrichment of cell fate commitment biological process from the DMRs associated with advanced prostate cancer may indicate the epigenetic regulation of lineage plasticity facilitating tumor adaptation and treatment resistance in advanced disease^{35,36}. Besides the biological processes, the transcriptional repressor activity and steroid hormone receptor activity enriched among the molecular functions suggest that epigenetic alterations in mCRPC may influence transcriptional dysregulation and androgen receptor signal which are both critical in driving prostate cancer progression and therapeutic resistance^{37–39}. Furthermore, the MHB-based composite risk score generated from the identified DMRs showed a significant association with mCRPC survival. We also observed that cfDNA methylation in the majority of target regions were correlated positively with predicted ctDNA fraction, suggesting that methylation could partially reflect tumor burden. However, we also identified a subset of regions which were not correlated or inversely correlated with ctDNA fraction. Interestingly, while the positively correlated and non-correlated MHBs had low MHL (< 0.05) at localized PC, the inversely correlated MHBs had elevated MHL (> 0.05) at localized PC which may be derived from tumor-associated cells such as immune cells in the tumor microenvironment. Finally, we integrated current prognostic clinical variables into the MHB-based risk score to develop patient-centric prognostic nomograms for individual mCRPC patients.

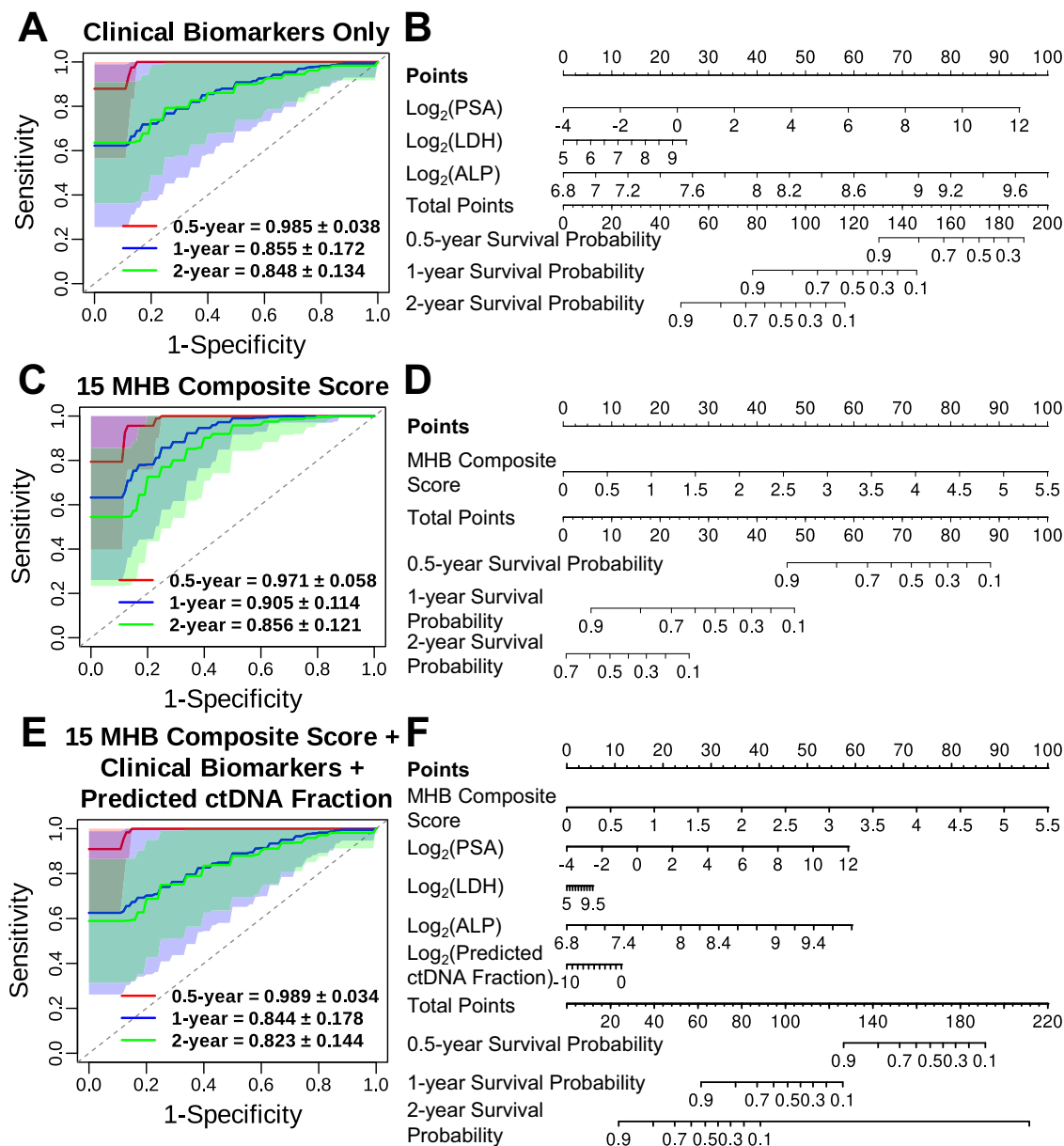


Fig. 5 | Nomograms built from clinical variables and MHBs predict probability of survival. **A** Time-dependent ROC AUC for 0.5-, 1-, and 2-year survival based on multivariable model utilizing PSA, LDH, and ALP. **B** Nomogram computing for 0.5-, 1-, and 2-year probability based on PSA, LDH, and ALP levels. **C** ROC AUC for model utilizing 15 MHB-based composite score. **D** Nomogram computing survival probability based on 15 MHB-based composite score. **E** ROC AUC for multivariable model utilizing clinical biomarkers, MHB-based composite score, and predicted

ctDNA fraction. **F** Nomogram based on clinical biomarkers, MHB-based composite score, and predicted ctDNA fraction. ROC curves represent the mean of 100× repeated 5-fold cross-validation, with shaded areas indicating ± 1 standard deviation. *HR* hazard ratio, *ROC AUC* Receiver Operating Characteristic Area Under the Curve, *PSA* prostate specific antigen, *ALP* alkaline phosphatase, *LDH* lactate dehydrogenase, *HR* hazard ratio, *CI* confidence interval.

Methylation changes occur across large genomic regions, often affecting multiple CpG sites within a locus and generating a stronger cumulative signal that enhances detection sensitivity and provides insights into tumor biology⁴⁰. These changes occur early in tumorigenesis^{41,42} and can change to reflect environmental and treatment exposures^{43,44}, while still expressing tissue-specific or tumor-specific patterns that allow for identification of the tissue of origin^{20,45}. This makes methylation profiling an attractive biomarker strategy particularly when ctDNA fractions are low during early-stage or minimal residual disease and fall below the detection limits of conventional ctDNA quantification methods, leading to false negatives. These quantification methods have been well reported using copy number variations (CNVs)^{46–48} and specific mutations^{49–51} but have limitations as they may lack sensitivity (CNV-

based) or require high sequencing depth to capture rare mutant alleles⁵² because of variability of mutations across tumor types and disease states⁵³. Furthermore, methylation patterns can capture inter-patient and intra-tumor heterogeneity that may not be reflected in ctDNA fraction alone, as samples with similar ctDNA levels can still exhibit distinct methylation profiles. Identifying methylation markers can therefore provide valuable insights into dynamic tumor biology, complementing established biomarkers to enhance risk stratification, guide therapeutic decisions, and uncover potential drug targets. This translational potential is exemplified by two regions included in the 15 MHB-based score—associated with *ALOX5* and *CDKN2A*—which are currently under investigation as direct or indirect targets for prostate cancer in preclinical studies and a clinical trial (NCT05113537)^{54,55}.

Exploring the potential for clinical translation, this study developed prognostic nomograms as patient-centric tools to predict mCRPC survival. Prognostic risk groups can classify patients into broad groups based on categorical variables by simplifying continuous data into categories. While this is helpful, it does not account for the unique combination of non-specific clinical variables present in each patient and potentially overlooks patient-level and tumor-biology based nuances. Unlike categorical risk-grouping, a nomogram-based approach handles categorical and continuous variables more efficiently. Nomograms offer a visual, individualized risk prediction approach that integrates multiple prognostic variables, making them more interpretable and clinically actionable than standard risk models^{56,57}. Although nomograms can be built using clinical variables alone, as indicated by their strong prognostic value in this study, the inclusion of tumor-specific methylation markers obtained through non-invasive blood draws enhances the model's prognostic accuracy. Furthermore, since clinical biomarkers such as ALP and LDH are not specific to tumor biology they lack the molecular context that tumor-specific methylation can provide. Methylation profiling can provide insights into potential disease mechanisms and inform treatment strategies.

The potential of cfDNA methylation as a critical biomarker in assessing treatment response and clinical outcomes cfDNA methylation in prostate cancer detection, prognosis, and monitoring^{58–61} has been reported. In one study dynamic methylated changes associated with disease progression and treatment response in metastatic prostate cancer⁵⁸ were observed. An independent study demonstrated that methylation patterns can serve as a proxy for tumor fraction in metastatic prostate cancer⁵⁹. A tumor tissue-informed approach showed that limited methylation markers are capable of differentiating treatment-responsive from non-responsive mCRPC patients and predicting progression-free survival in mCRPC patients⁶⁰. Lastly, a study employing a genome-wide approach identified methylation signatures that could distinguish between localized and metastatic cancer. Furthermore, methylation level at a gene promoter region was associated with progression-free survival and overall survival⁶¹. Building on these findings, this study incorporated cfDNA methylation into an integrative prognostic model to explore a tumor biology-based patient-centric application for cfDNA methylation in mCRPC prognostication. Going beyond traditional non-specific clinical variables, cfDNA methylation reflects altered tumor biology-based features that may provide future opportunities for therapeutic targeting in the mCRPC state.

There are several advantages to using a targeted rather than a whole genome methylation sequencing approach. First, the targeted EM-seq is more sensitive, cost-effective, and avoids DNA degradation associated with bisulfite-based sequencing approaches^{62,63}. This enhanced sensitivity allows for the detection of subtle methylation differences that may be missed by traditional methods. Second, prior studies have typically analyzed mean methylation levels across sequencing reads within predefined genomic windows or specific target regions^{59,60,64}. This approach may overlook the heterogeneity of methylation patterns within individual DNA fragments, further masking variations in methylation haplotypes that may have functional relevance in cancer progression. In this study, we leveraged MHBs which captured the coordinated methylation patterns across multiple CpGs within the same DNA fragment²³ and in different states of cancer progression. By considering read-level methylation haplotypes and including samples collected from different progressive cancer states, a more granular view of regions undergoing tumor-specific epigenetic alterations in mCRPC could be obtained since the MHBs also consider interactions between CpG sites. Third, we generated a patient-centric prognostic nomogram, a methodology which has not commonly employed in prior cfDNA methylation studies, providing an intuitive, clinically actionable framework for survival prediction.

This study highlights promising findings as a proof-of-principle but has several limitations. The small sample size limits the generalizability of the results, and future validation in larger, independent cohorts is necessary to confirm the study approach and the robustness of the prognostic models. Additionally, the detection of methylation signals in

cfDNA may be influenced by ctDNA levels, as regions with low ctDNA abundance may fall below the detection threshold, potentially biasing the observed methylation patterns toward those more readily detectable in advanced disease. While we attempted to account for this by analyzing the correlation between methylation and predicted ctDNA fraction, this study relied on predicted rather than directly measured ctDNA levels, which may introduce uncertainty. A direct ctDNA quantification alongside methylation profiling could provide a better understanding of their relationship. Nonetheless, the findings in this study highlight the potential of cfDNA methylation profiling as a clinically relevant tool for prognostication and disease monitoring in prostate cancer. These cfDNA methylation signatures could complement existing prognostic models to offer tumor-biology-based prognostic tools. Since cfDNA-based testing is non-invasive, scalable, and allows for longitudinal monitoring without the need for tissue biopsies, if validated this approach will aid in the development of patient-centric, biology-informed nomograms for individualized risk stratification.

Methods

Patient cohort

This study used plasma samples from a prospective, clinically annotated, real-world patient biobank. To avoid preanalytical variations, the sample collection followed uniform standard operating procedures (SOPs). All SOPs include a uniform blood-based collection of specimens in patients visiting Huntsman Cancer Institute (HCI). Prostate cancer patients receiving treatments are approached under institutional biospecimen collection Institutional Review Board (IRB)-approved study of “Total Cancer Care” (IRB# 00089989; IRB# 00139755) and written informed consent is obtained. In addition, the study (#MCC 20563) also received approval by the Moffitt Cancer Center IRB. Collection of blood is performed under Clinical Laboratory Improvement Amendments (CLIA) supervision of the Biorepository and Molecular Pathology (BMP) Core facility at HCI and routed to the investigator's research lab for uniform processing. All collections used in this study are performed in EDTA tubes and plasma generated from platelet-poor fractions used for proteomic analyses and processing is completed within two hours of collection and banked in -80°C and with no freeze-thaws. Granular clinical annotation is obtained from EMRs with data dictionaries.

Blood collection and cfDNA extraction

Whole blood collected in 4.5 mL EDTA tubes from all patients in the biobank was subjected to centrifugation at 2000 rpm for 10 min and a second centrifugation of the supernatant at 3000 rpm for 10 min to make platelet-poor plasma. 0.5–1 mL fractions of the plasma samples were stored at -80°C until use. 1–2 mL of plasma was used for automated cfDNA extraction with the Maxwell RSC ccDNA Plasma Kit (Promega, #AS1480). cfDNA quantification was performed using Qubit.

Capture probe design

437 targeted methylation regions were selected from three major sources, including (1) Literature search identified 184 DMRs associated with prostate cancer^{65–69}; (2) TCGA-PRAD²⁰ data showed 103 DMRs when compared to normal controls. The selection criteria used were a difference in mean methylation β value of >0.4 between prostate tumor tissues and normal tissues, and a mean methylation β value <0.05 in plasma cfDNA from healthy controls; and (3) Prostate cancer-related oncogenes and tumor suppressor genes ($n = 79$) (Supplementary Data 2). We also included 29 positive and 42 negative control regions. The positive and negative control regions were selected based on regions that had consistently shown $>95\%$ methylation level and $<0.5\%$ methylation level in cfDNA from healthy controls²⁰, respectively. Overall, we selected 437 targeted regions, including 366 prostate cancer-related regions, 29 positive control, and 42 negative control regions. The selected regions covered all CpG islands of targeted genes and $\pm 5\text{Kb}$ flanking sequences. All capture probes were designed by Twist Custom Design Service.

EM-seq library preparation and target capture

10–20 ng of cfDNA was subjected to Enzymatic Methyl-seq (EM-seq) (NEB, #E7120S). Briefly, the cfDNA was first ligated with methylated adaptors, followed by oxidation using TET2 reaction and deamination using APOBEC reaction. Indexed primers were then used for 8–10 cycles of amplification. To enrich regions of interest, whole genome sequencing libraries were captured with a custom-made Twist targeted methylome panel covering 3.44 Mbp of genome. The captured libraries were further amplified for an additional 5 cycles before performing 150 bp paired-end (PE) sequencing in an Illumina sequencer.

Data preprocessing and methylation calling

Quality control checks were performed with fastqc (v0.11.9) and multiqc (v1.12) on the FASTQ files. Illumina adapter sequences were trimmed with cutadapt (v2.8), with a filter applied to discard reads shorter than 20 bp. Alignment to the human reference genome hg38/GRCh38 and methylation calling were performed on the processed reads using Bismark (v0.22.3) with bowtie2 (v2.3.5.1)²¹. Alignment parameters were set to a multi-seed length of 22 bp with 1 mismatch (-L 11, -N 1). Deduplication to remove alignments at the same position in the mapped genome was performed with Bismark (deduplicate_bismark). Mappable reads and duplicate alignments were extracted from Bismark alignment reports. To evaluate sequencing coverage in target regions, mosdepth (v0.3.3) was utilized to generate coverage profiles for each sample.

Methylation haplotype blocks (MHBs) identification and methylation haplotype load (MHL) analysis

Methylation haplotypes were extracted by converting deduplicated BAM files into mHAP format with mHapSuite (v2.1) (convert), available at <https://github.com/yoyoong/mHapSuite>²². MHBs were then identified with mHapSuite (MHBDiscovery) within the target regions (± 5 Kb) across all patients, based on the parameters of a core window of 3, and r^2 cutoff of 0.3, and p -value cutoff of 0.05. MHL values were determined by calculating the methylation statistics of MHBs using mHapSuite (stat). MHL determines the level of consecutive CpG methylation at different lengths. MHL is calculated as:

$$MHL = \frac{\sum_{i=1}^{10} i \times P(MH_i)}{\sum_{i=1}^{10} i} \quad (1)$$

Where $P(MH_i)$ is the fraction of consecutively methylated CpGs at length i . For haplotypes with more than 10 consecutive CpGs, all lengths from 1 to 10 are considered. Identified MHBs were further filtered to retain only those with greater than 3 CpG sites and greater than 50 median reads across all patients. MHBs that overlapped with positive and negative control regions were excluded from subsequent analyses for determining differential MHBs.

Identification of differential MHBs

Prior to identifying which MHBs are differentially methylated, MHBs missing MHL values were excluded. Welch's t -tests were performed for the comparison of MHBs between patient cohorts (localized PC vs. mHSPC, and mHSPC vs. mCRPC). MHBs where the mean MHL values were less than 0.05 in localized PC patients were used for subsequent comparative analyses. MHBs with a $p < 0.05$ were considered as DMRs. Persistent DMRs present in both comparisons (localized PC vs. mHSPC, and mHSPC vs. mCRPC) were identified with a $p < 0.05$ and FDR < 0.1 through the Benjamini-Hochberg (BH) procedure. Volcano plots displaying the mean differences between localized PC vs. mHSPC and mHSPC vs. mCRPC were created using the tidyverse (v2.0.0) R package. The Venn diagrams detailing the overlaps between the two comparisons were created using the Venn-Diagram (v1.7.3) R package.

Survival analyses

Survival probabilities for each MHB were assessed using multiple univariate Cox proportional hazards models with the survival (v3.5.8) R package.

A reduced panel of MHBs was selected based on the top statistically differentially methylated MHBs, which overlapped in all t -test comparisons and had Cox model $p < 0.05$. When multiple MHBs represented the same gene region, the MHB with the highest hazard ratio was selected for the panel. A composite score (C) for the selected MHB panel was determined using leave-one-out cross validation (LOOCV). For each iteration ($n = 49$) the composite risk score was calculated for the training set, and the median composite risk score of the training set was used as the threshold to classify composite risk score from the remaining test set as high-risk ($>$ training median) or low-risk ($<$ training median). The composite risk score (C) was calculated as the sum of the products of the univariate Cox coefficient (β) and the corresponding MHL value (M) for each MHB: (2) $C = \beta_1 M_1 + \beta_2 M_2 \dots$.

Another composite risk score (B) for significant clinical biomarkers (PSA, ALP, LDH) was calculated as the sum of the products of the multi-variable Cox coefficient for each clinical biomarker and the corresponding measurement of the clinical biomarker (V) for each patient (3) $B = \beta_1 V_1 + \beta_2 V_2 + \beta_3 V_3$.

Integrated risk scores (R) incorporating the MHB composite score (C) and selected clinical biomarkers (V) were calculated by multiplying the multivariable Cox coefficient for each covariate with its associated value and summing the resulting products (4) $R = \beta_1 C_1 + \beta_1 V_1 + \beta_2 V_2 \dots$.

The added prognostic value of the MHB composite score relative to predicted ctDNA fraction was assessed by comparing Cox proportional hazards models based on ctDNA fraction alone, and ctDNA fraction and MHB composite score. Model comparison was performed with nested analysis of variance (ANOVA) with likelihood ratio test (LRT). Statistical significance was determined based on the LRT statistic and $p < 0.05$.

For survival models (B) and (R), risk scores which were greater than the median risk score were categorized as high-risk while risk scores lower than the median risk score were categorized as low-risk. Clinical biomarkers were log-transformed by a base of 2 for survival analyses. Differences in survival probability were evaluated with Wald test, with a $p < 0.05$ considered to be statistically significant. Kaplan-Meier plots to represent the differences in survival probability were generated using the survminer (v0.4.9) R package.

Prediction of ctDNA fraction and correlation with MHL

Predicted plasma ctDNA fractions for mCRPC patients ($n = 49$) were determined by inputting cfDNA, PSA, LDH, ALP, ECOG Performance Score, liver metastasis, and lung metastasis information for each patient into the online tool <https://www.ctdna.org>²⁴. Patients with $< 1\%$ predicted ctDNA fraction ($n = 14$) were excluded from correlation analyses. Among the remaining mCRPC patients ($n = 35$) Pearson correlation was performed between the predicted ctDNA fraction and the 15 MHB-based composite score, as well as with the MHL of each MHB that had no missing values. MHBs with Pearson $r > 0.5$ were identified as positively correlated with predicted ctDNA fraction. MHBs with Pearson $r > -0.2$ and < 0.2 were identified as not correlated with predicted ctDNA fraction. MHBs with Pearson $r < 0$ were identified as inversely correlated with predicted ctDNA fraction. The plots depicting the correlation between ctDNA fraction and MHBs were generated with the tidyverse and ggpubr (v0.6.0) R package.

Nomogram development

To estimate individual survival probabilities, survival nomograms were built based on univariate and multivariable Cox proportional hazards regression models. These models quantified the relationship between survival outcomes and multiple predictors. The Cox coefficients derived from each model were used to assign point values, allowing for the calculation of a total score that reflects an individual's overall survival risk profile. Model performance was evaluated using 100x repeated 5-fold cross validation. Mean time-dependent Receiver Operating Characteristic Area on the Curve (ROC AUC) and standard deviation were computed across repetitions.

The nomograms were developed based on the identified predictors for predicting the 0.5-, 1- and 2-year survival, and each predictor was represented on a separate axis in the nomogram. 0.5-, 1- or 2-year survival

probability is acquired by calculating the total score of all predictors. For a given predictor, the number of points can be determined by drawing a vertical line from the measured predictor value to the ‘points’ scale at the top of the nomogram. The sum of the points from each predictor is equal to the total points score. After determining the total points score, survival probability can be determined by drawing a vertical line from the total points score in the ‘total points’ scale to the survival probability scales at the bottom of the nomogram. Intersection with a survival probability scale (0.5-, 1-, 2-year) indicates the patient’s predicted probability of survival (0–0.9). Nomograms were constructed with the rms (6.8.0) R package. Time-dependent ROC curves were determined and plotted with the timeROC R (0.4) package.

Data availability

De-identified sequencing data from patients in this study is available on request. Processed methylation data for all samples are provided in the Supplementary Data.

Code availability

Scripts for reproducing tables and figures in the manuscript are available in the following GitHub repository (<https://github.com/jodie-wong/mCRPC-prognosis-manuscript>).

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Competing interests

The authors declare no competing interests.

Additional information

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